The two rational sign projectors

An integral involution separates over the rationals with denominator 2.

$$E_+ = (I + A)/2 \qquad E_- = (I - A)/2$$

$$\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

$$E_+^2 = E_+, \quad E_-^2 = E_-, \quad E_+ E_- = E_- E_+ = 0$$

Here A = [[1, 1], [0, −1]]. Both projectors have exact 2-adic denominator weight one.